

Quantifiers:-

$X = \{x_1, x_2, x_3, \dots\}$ infinite

show that every $x_n \in X$ is odd.

→ $P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge P(x_4) \wedge \dots$
 $\forall x \in X, P(x)$

show that at least one $x_n \in X$ is odd.

→ $P(x_1) \vee P(x_2) \vee P(x_3) \vee P(x_4) \vee \dots$
 $\exists x \in X, P(x)$

$\forall$ for all

$\exists$ there exists

universal quantifier

existential quantifier

ex:- for every $n \in \mathbb{Z}$, $2n$ is even

$$\forall n \in \mathbb{Z}, 2n \text{ is even}$$

$$\forall n \in \mathbb{Z}, \underline{E(2n)}$$

where $E(k)$ tells us k is even.

ex:- There exists a subset of $\mathbb{N}$ for $|x| = 5$

$\exists$

$\subseteq$

$$\exists x \subseteq \mathbb{N}, |x| = 5$$

$$\exists x \in \mathcal{P}(\mathbb{N}), |x| = 5$$

$$\exists x, (x \subseteq \mathbb{N}) \wedge (|x| = 5)$$